Manifold-Aligned Consistency Contrastive Learning for Noise-Tolerant Hyperspectral Image Classification

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***Abstract*—Label noise in hyperspectral image classification (HIC) has posed a significant challenge, mainly due to the complexity of high-dimensional spectral–spatial data. Existing approaches that rely on sample predictions for verification and correction often amplify confirmation bias, leading to decision boundaries that overfit to noisy labels. To address these issues, we propose a manifold-aligned consistency contrastive learning (MACCL) framework that establishes a mutually guided consistency alignment mechanism between the spectral–spatial representation space and label space through manifold learning theory to combat label noise. Specifically, to mitigate the confirmation bias of noisy labels, we introduce a manifold-aligned consistency learning module. It leverages the manifold assumption in the representation space, modeling local neighborhood graphs and enforcing consistent prediction distributions via KL divergence. This aligns label-space classifications with the representation space's local geometry, suppressing isolated noise through manifold continuity. In addition, to combat representation degradation that causes decision boundaries to overfit noisy labels, we integrate a noise-tolerant contrastive representation learning (NTCRL) module. By applying confidence-guided criteria, the module focuses on high-confidence sample pairs and regularizes gradients. This emphasizes clean pairs during training, boosting the model's discriminative ability, and preserving true semantic relationships. Through this mutual guidance, the contrastive learning (CL) refines the local geometric structure through discriminative representation learning, driving the representation space closer to the intrinsic data manifold, while the manifold alignment propagates geometric constraints to rectify label-space corruptions. Finally, experiments on several benchmark datasets with varying levels of noise have validated the superiority of the proposed MACCL framework.**

***Index Terms*—Confidence guided, contrastive learning (CL), hyperspectral image classification (HIC), manifold alignment, noisy labels.**

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I. Introduction

HYPERSPECTRAL images (HSIs) meticulously capture data across hundreds to thousands of contiguous spectral bands, thereby acquiring spatial and spectral information that substantially transcends the limitations of traditional RGB imagery. This acquisition of highly detailed spectral information renders HSI classification (HIC) a potent and indispensable analytical technique that enables precise identification and classification of surface materials by leveraging their intricate spectral characteristics, particularly in domains such as agriculture, geological exploration, and environmental monitoring [1], [2], [3]. In the field of HIC, significant progress has been made, with notable advancements in algorithms and data processing techniques. However, despite these significant achievements, the prevalence of noisy labels remains a persistent challenge, severely affecting the precision, robustness, and general reliability of classification models in this domain [4], [5], [6], [7]. Noisy labels are inaccurate or erroneous annotations arising from human mislabeling, the imprecision of automated processes, inconsistent expert interpretations, or discrepancies between assigned labels and actual categories due to the intrinsic characteristics of the samples. In practical applications, noisy labels are often inevitable, and deep networks are prone to overfitting these noisy labels, leading to a drastic decline in generalization performance [8], particularly in supervised learning scenarios where the accuracy of labeling is crucial for training sophisticated classifiers [9], [10]. Consequently, the identification and mitigation of noisy labels in HIC are critical and necessary endeavors, aimed at improving the overall reliability, robustness, generalization capability, and accuracy of the classification process in real-world applications.

Learning with noisy labels (LNL) has been initially highlighted in computer vision, especially for image classification [11], [12], [13], [14]. However, these methods are not directly applicable to HIC due to complex spectral–spatial patterns and high-dimensional feature spaces with significant intraclass variability and interclass similarity [15], [16]. Recently, LNL in HIC has drawn considerable attention, which can be summarized into two main research approaches.

1) *Data-Driven Noisy Labels Handling:* Data-driven approaches focus on identifying and correcting mislabeled samples. Specifically, noisy samples are identified

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via spectral–spatial similarity [17], density estimation [18], and related techniques, after which their labels are rectified [19], reweighted, or corrected through label propagation [20].

2) *Model-Driven Noisy Labels Handling:* Model-driven approaches improve robustness to noisy labels through robust loss functions or architectural innovations. By designing robust loss functions [15], refining network architectures [21], or incorporating contrastive representation learning mechanisms [22], model-driven methods allow end-to-end training procedures to intrinsically suppress the detrimental effects of noisy labels.

Recent studies have shown that contrastive learning (CL) [23], [24] helps to learn significant discriminative representations and improves downstream task performance. Based on the availability of supervision, it can be divided into self-supervised CL [25] and supervised learning [26]. As shown in Fig. 1 (left), self-supervised CL pushes same-class samples farther apart, deviating from the classification task objective. In contrast, supervised CL constructs correct positive pairs from the same-class samples in classification tasks. However, if the supervision is corrupted by noisy labels, the constructed pairs become noisy, leading to incorrect representation learning. As shown in Fig. 1 (middle), erroneous supervision causes category mismatches. Inspired by this, pseudo-labeling redefines contrastive pairs to reduce adverse effects [27], enabling robust representation learning. However, the unreliability of pseudo-labels and the inevitable erroneous pairs can dominate the representation learning, resulting in suboptimal generalization performance.

Despite the achievements in the field of HIC with noisy labels, two significant limitations persist.

1) *Propagation of Confirmation Bias:* With noisy labels, the model may rely on its predictions during training, reinforcing incorrect labels. As shown in Fig. 2 (left), the model overfits to noisy labels, resulting in decision boundaries that reflect these biases. Once erroneous samples are incorporated, the model perpetuates these biases, causing distorted predictions in subsequent steps [28].
2) *Manifold Misalignment and Feature Degradation:* Existing methods often fail to fully account for the impact of noisy labels on the intrinsic geometric structure of the data manifold [29]. As a result, the learned feature representations become misaligned, impairing the model's ability to distinguish between clean and noisy examples effectively, and underutilizing the potential of robust features that can separate them [30].

In recent years, manifold learning has primarily focused on preserving the intrinsic neighborhood structures within low-dimensional feature spaces. Classical manifold learning techniques, such as LLE [31], estimate local manifolds through well-founded geometric assumptions. Furthermore, the consistency hypothesis in semi-supervised learning [32] suggests that samples sharing local manifold structures are likely to possess identical labels. Building upon these insights, we propose the manifold-aligned consistency CL (MACCL) framework, which pioneers the integration of manifold geometric consistency

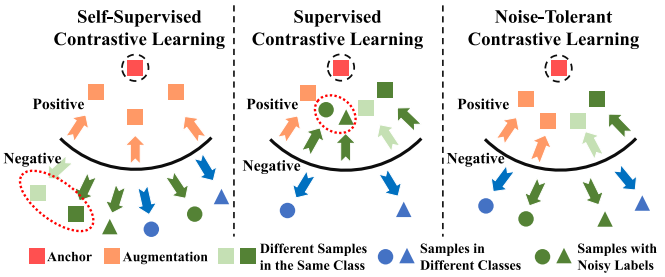


Fig. 1. Illustration of CL approaches. (Left) Self-supervised CL pushes the same-class samples apart. (Middle) Supervised CL pulls noisy nonsimilar samples closer. (Right) Our noise-tolerant CL effectively addresses this by selecting reliable pairs and improving representation alignment.

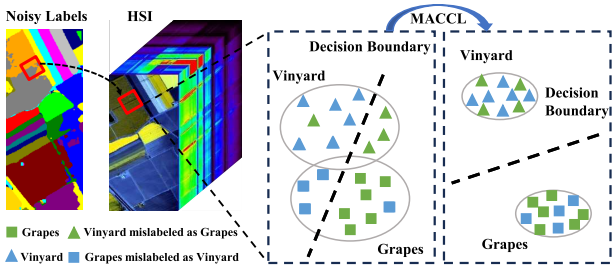


Fig. 2. Illustration of HSI classification decision boundaries with noisy labels. Black curves represent the decision boundaries learned during training. (Left) Previous methods, where the learned decision boundary overfits the noisy labels, and the representation space is scattered. (Right) Our approach, where the decision boundary is robust to noise, aligns well with the data distribution, and the learned representation manifold is compact.

constraints with noise-tolerant CL. This innovation establishes a mutually guided consistency alignment mechanism between the representation space and label space to block the propagation of confirmation bias and enhance cluster compactness. As illustrated in Fig. 2 (right), this mechanism forces the classifier's decision boundaries to align with the intrinsic structure of the data manifold. Our main contributions are as follows.

1) The MACCL framework establishes a mutually guided consistency alignment between spectral–spatial representation space and label space through manifold learning. By integrating manifold-aligned consistency with CL, MACCL mitigates confirmation bias propagation while enhancing cluster compactness, enabling noise-tolerant HIC.
2) To mitigate confirmation bias, we develop a novel manifold-aligned consistency module. It explicitly models local geometric structures through neighborhood graphs and enforces prediction consistency via KL divergence constraints. This aligns label-space decisions with the intrinsic data manifold's continuity prior, effectively suppressing isolated noisy labels through structural propagation while preserving authentic category relationships.
3) To combat representation degradation that causes decision boundaries to overfit noisy labels, we introduce a noise-tolerant CL module. It uses confidence-guided criteria to focus on high-confidence sample pairs and regularizes gradients, emphasizing clean pairs to boost the model's discriminative ability.

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The subsequent sections of this article are organized in the following manner. An overview of related research is provided in Section II. The architecture and specifics of the proposed framework are elaborated in Section III. Section IV is dedicated to discussing the findings from various experiments along with a comprehensive analysis. Finally, this article is summarized with concluding remarks in Section V.

II. Related Works

A. Learning With Noisy Labels

In the realm of noisy labels learning within computer vision, the predominant approaches can be classified into four principal strategies: noise transition matrix [33], robust loss functions [34], sample selection, and label correction [35], [36] strategies. Early approaches aimed to rectify losses by introducing a noise transition matrix. However, estimating these matrices poses challenges, necessitating prior knowledge or well-annotated data subsets. Other methods designed noise-robust loss functions to rectify losses based on network predictions. Nonetheless, these techniques tend to falter under high noise levels. Recently, co-training-based sample selection and label correction algorithms have garnered widespread attention, with the core principle of identifying clean data and removing or rectifying noisy instances. Illustratively, methodologies such as co-teaching [35] employ dual networks during the training process, wherein each network selectively learns from the low-loss samples identified by its counterpart. Similarly, DivideMix [37] employs a Gaussian mixture model (GMM) on the losses of two networks for mutual sample selection and applies semi-supervised learning techniques, aiming to rectify noisy labels based on the averaged predictions from different data augmentations. However, due to the properties of HSIs, computer vision models are inapplicable to HIC tasks.

Recently, LNL in HIC has drawn considerable attention. Early methods focused on detection and removal. For example, Tu et al. [17] presented the spectral angle and the local outlier factor (SALOF) algorithm for noisy label detection in HIC. Tu et al. [18] amalgamated superpixel-to-pixel weighting distance with density peak clustering to effectively identify and remove noisy labels. Recognizing the risk of information loss from discarding noisy labels, contemporary studies focus on detecting and correcting mislabeled data. Kang et al. [19] combined domain transform recursive filtering with constrained energy minimization to correct mislabeled samples. Xu et al. [15] presented the dual-channel residual network (DCRN), featuring an innovative structure and a noise-resistant loss function. Zhang et al. [22] presented a triple contrastive representation learning (TCRL) framework from a deep clustering perspective for robust HIC. However, reliance on individual predictions for selection and correction in these methods can potentially lead to confirmation bias. Therefore, distinct from existing approaches, we propose the MACCL framework to address noisy labels in HIC by explicitly modeling local geometric structures through manifold alignment, propagating structural priors to suppress confirmation bias, and enhancing robustness.

B. HIC via Deep Neural Network

As deep learning technology advances, a myriad of effective models has been developed to tackle the classification challenges of HSI. Notably, Qin et al. [38] proposed a language-enhanced dual-level CL network (LDCLNet) for open-set HIC, using textual supervision to improve recognition of unknown classes. Similarly, Qin et al. [39] proposed a novel domain generalization network with frequency disentanglement and data geometry (FDGNet) for cross-scene HIC. Feng et al. [40] proposed a cross-domain few-shot learning based on decoupled knowledge distillation (DKD-FSL) for HIC, which integrates metaknowledge extraction and debiasing through a decoupled logit interaction module. Wang et al. [41] proposed a multistage information complementary fusion network based on flexible-mixup (Flex-MCFNet), which enhances classification by integrating hyperspectral and multimodal data. Wang et al. [42] proposed a representation-enhanced status replay network (RSRNet), which employs dual augmentation, status replay strategy, and cross-modal interactive fusion to enhance HIC. Hu et al. [43] proposed a global–local self-attention network (GLSANet), which combines global atrous and local window self-attention mechanisms for remote sensing image segmentation. Zhang et al. [44], [45] proposed both a single-source domain expansion network (SDEnet) employing encoder–randomization–decoder architecture with adversarial learning for cross-scene HIC and a bidirectional domain adaptation (BiDA) framework using a triple-branch transformer for cross-domain HIC. To extract crucial information and mitigate the impact of noise and redundancy in high-dimensional hyperspectral imagery, numerous attention mechanisms are employed in the domain of HIC. For example, Hong et al. [46] introduced SpectralFormer, a Transformer-based model for spectral extraction, and Zhu et al. [47] developed a residual spectral–spatial attention network.

Although deep learning strategies have achieved successes in HIC, these methods often rely on a substantial volume of accurately annotated data. In practice, datasets often contain mislabeled samples, degrading performance [9]. To enhance the robustness of models against noisy labels in HIC, researchers have attempted to replace cross-entropy loss (CELoss) with noise-resistant loss functions [6], [15], aiming to alleviate overfitting issues. However, these methods conduct robust learning through their predictions, leading to confirmation bias and still ultimately overfitting the noisy labels.

C. Contrastive Learning

CL [23], [24], [26] constitutes a foundational methodology in self-supervised representation learning. Specifically, SimCLR [23] establishes cross-view invariance through systematic integration of multimodal augmentations and a redesigned InfoNCE loss [25]. SwAV [48] optimizes computational efficiency via online clustering with synchronized prototype consistency. SimSiam [24] redefines contrastive paradigms by achieving stable feature learning solely through stop-gradient operations on symmetrized positive pairs. Supervised variants like SupCon [26] enhance class separability via label-driven contrastive objectives, yet their dependency on annotation quality limits scalability.

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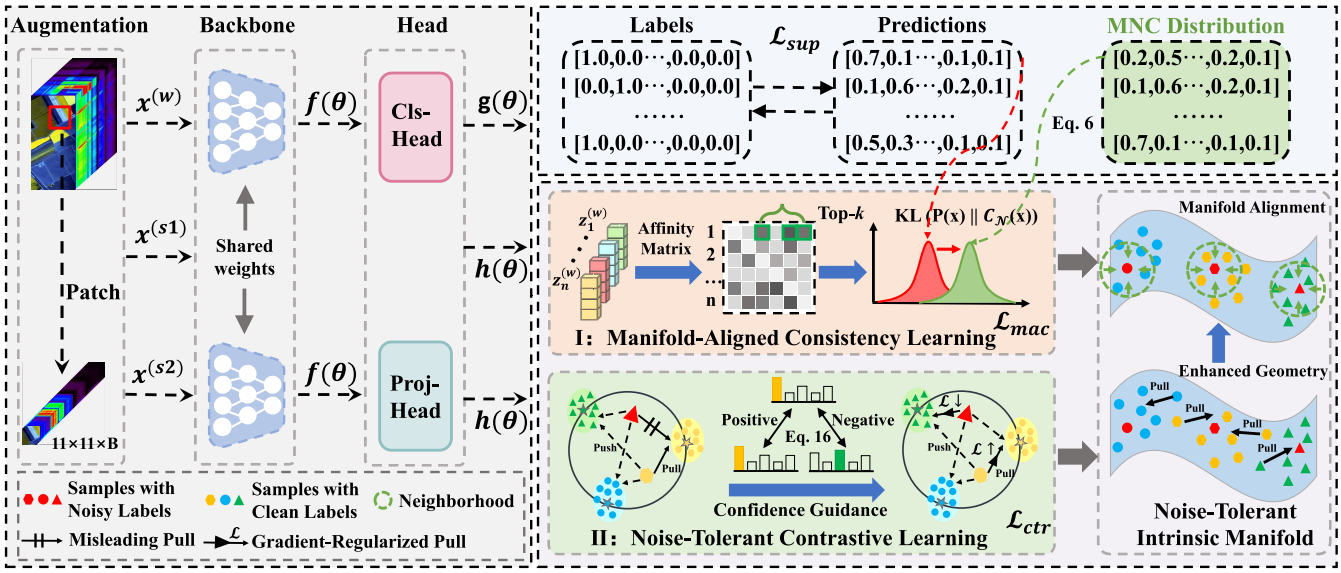


Fig. 3. Illustration of the proposed MACCL framework. This architecture combines data augmentation, a shared backbone for feature extraction, and two heads: the classification head for label prediction and the projection head for representation learning. It incorporates manifold-aligned consistency learning to align label predictions with the data manifold and leverages a noise-tolerant contrastive module that employs confidence-guided pair selection and gradient regularization to refine the spectral–spatial representation space, mitigating the impact of noisy labels for robust HIC.

III. Proposed Method

A. Problem Formulation

Given hyperspectral data are formalized as $\mathbb{H} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B denote the dimensions corresponding to height, width, and spectral bands, respectively. To effectively utilize both spectral and spatial information, each pixel is transformed into a compact spectral–spatial cube represented as $\boldsymbol{x}_i \in \mathbb{R}^{L\times L\times B}$, with L indicating the edge length of the cube. The training dataset is defined as $\mathcal{D} = \{(\boldsymbol{x}_i, \boldsymbol{y}_i)\}_{i=1}^{n}$, where $\boldsymbol{y}_i \in \{0,1\}^c$ formatted as one-hot encoded labels. The objective in HIC is to optimize the mapping function $\mathcal{F}: \boldsymbol{x} \to \boldsymbol{y}$, controlled by a designated loss function $\mathcal{L}$, aiming to substantially reduce the empirical risk, denoted as $\mathcal{R}_{\mathcal{L}}(\mathcal{F})$

$$\mathcal{R}_{\mathcal{L}}(\mathcal{F}) = \mathbb{E}_{\mathcal{D}}\left[\mathcal{L}(\mathcal{F}(\boldsymbol{x}), \boldsymbol{y})\right] = \frac{1}{|\mathcal{D}|}\sum_{(\boldsymbol{x},\boldsymbol{y})\in\mathcal{D}} \mathcal{L}(\mathcal{F}(\boldsymbol{x}), \boldsymbol{y}). \quad (1)$$

In the presence of noisy labels, the training process incorporates a dataset $\widehat{\mathcal{D}} = \{(\boldsymbol{x}_i, \hat{\boldsymbol{y}}_i)\}_{i=1}^{n}$, where $\hat{\boldsymbol{y}}_i$ denotes the noisy label for the ith sample, and $\hat{\boldsymbol{y}}_i$ does not necessarily equal to $\boldsymbol{y}_i$. Consequently, during training epoch t, a mini-batch $\mathcal{B} = \{(\boldsymbol{x}_i, \hat{\boldsymbol{y}}_i)\}_{i=1}^{b}$ is randomly selected from $\widehat{\mathcal{D}}$, with b representing the batch size. The model parameters are updated based on the gradients computed from this mini-batch $\mathcal{B}_t$, a step that ensures the iterative refinement of the model parameters throughout the training process, expressed as

$$\Theta_{t+1} = \Theta_t - \eta\nabla\left(\frac{1}{|\mathcal{B}|}\sum_{(\boldsymbol{x},\hat{\boldsymbol{y}})\in\mathcal{B}} \mathcal{L}(\mathcal{F}(\boldsymbol{x};\Theta_t), \hat{\boldsymbol{y}})\right) \quad (2)$$

where η is the learning rate and Θ represents the trainable parameters of the model.

However, noisy labels may distort gradients, skewing the data distribution and leading to erroneous parameter updates. This misdirection may induce the model to adapt excessively to noisy labels, thereby ingraining inaccuracies and compromising overall validity. Thus, developing robust methods to mitigate noisy labels is vital for improving HIC reliability.

B. Overall Framework of MACCL

In the given framework, each batch of samples is represented as $\mathcal{B} = \{(\boldsymbol{x}_i, \hat{\boldsymbol{y}}_i)\}_{i=1}^{b}$, where $\boldsymbol{x}_i$ indicates the input and $\hat{\boldsymbol{y}}_i$ indicates the corresponding label for the ith sample. Each input $\boldsymbol{x}_i$ undergoes weak data augmentation to generate $\boldsymbol{x}_i^{(w)}$ and strong data augmentation to produce two variants, $\{\boldsymbol{x}_i^{(s_k)}\}_{k=1,2}$. These augmented inputs are then processed by a feature extractor f. Following the extraction, the resulting feature vectors are fed into two distinct neural network heads: a classification head g and a projection head h. The classification head g computes the classification probabilities $p(\boldsymbol{x}_i) = g(f(\boldsymbol{x}_i^{(w)}))$ for the weakly augmented inputs. Simultaneously, the projection head h outputs feature mappings from both weakly and strongly augmented inputs, denoted as $\boldsymbol{z}_i^{(w)} = h(f(\boldsymbol{x}_i^{(w)}))$, $\boldsymbol{z}_i^{(s_k)} = h(f(\boldsymbol{x}_i^{(s_k)}))$, respectively.

As depicted in Fig. 3, the MACCL framework comprises three core modules designed to combat noisy labels in HIC.

1) *Manifold-Aligned Consistency Learning:* This module enforces label consistency by exploiting the local manifold structure of the data. It aligns predictions with the geometric continuity of neighboring samples, suppressing isolated noisy labels through neighborhood-based propagation. This ensures robustness by anchoring labels to the intrinsic data topology rather than corrupted annotations.
2) *Noise-Tolerant Contrastive Representation Learning (NTCRL):* This module introduces a confidence-guided pair selection strategy to avoid erroneous contrastive

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pairs induced by noisy labels. By amplifying gradients from semantically correct pairs and suppressing noisy ones, it learns discriminative representations while resisting contamination from noisy labels.

3) *MixUp-Based Supervised Learning:* Linear interpolation of samples and labels smooths decision boundaries, reducing overfitting to noisy instances. This implicit regularization enhances generalization by emphasizing global data distribution over local label noise.

These modules synergistically counteract confirmation bias caused by noisy labels: CL refines feature spaces, manifold alignment propagates geometric constraints, and MixUp stabilizes training. Together, they ensure robust HIC under varying noise levels through theoretical guarantees of error bounds and gradient dominance. Sections III-C–III-E introduce the key components in detail.

C. Manifold-Aligned Consistency Learning

According to the manifold hypothesis [32]: samples sharing local manifold structures are likely to possess identical labels, we design a manifold-aligned consistency constraint leveraging neighborhood relationships on the data manifold. Our method enforces local prediction consistency through geometric constraints derived from the manifold's continuity prior, where samples sharing intrinsic neighborhoods are regularized to exhibit aligned label distributions. This mechanism inherently suppresses label noise violating local geometric coherence while preserving authentic category boundaries. The theoretical analysis further establishes error bounds proportional to the deviation between label assignments and manifold geometry, formally grounding MACCL's noise robustness.

Definition 1 (Manifold Neighborhood Confidence (MNC)): Let $\mathcal{D}$ be a dataset lying on a low-dimensional manifold $\mathcal{M}$. For a sample $\boldsymbol{x}_i \in \mathcal{D}$, the manifold neighborhood $\mathcal{N}(\boldsymbol{x}_i)$ as the set of samples in $\mathcal{D}$ that are geometrically proximate on $\mathcal{M}$. The MNC $C_{\mathcal{N}}(\boldsymbol{x}_i; \mathcal{D})$ is

$$C_{\mathcal{N}}(\boldsymbol{x}_i; \mathcal{D}) = \sum_{\boldsymbol{x}_j \in D} \mathbf{1}\left[\boldsymbol{x}_j \in \mathcal{N}(\boldsymbol{x}_i)\right] \cdot C\left(\boldsymbol{x}_j\right) \tag{3}$$

where $C(\boldsymbol{x}_j)$ is the prediction confidence of example $\boldsymbol{x}_j$.

Assumption 1 (Expansion and Separation): Following the assumption [49], the α-expansion and β-separation assumptions hold for the training set $\mathcal{D}$. The α-expansion means that a sample is reachable to the α number of augmentation neighbors on average, i.e., $\mathbb{E}_{x \in \tilde{\mathcal{D}}}[|\mathcal{N}(\boldsymbol{x})|] = \alpha$. The β-separation means that data distributions with different ground-truth classes are highly separable, with the average proportion of the neighbors from different classes being as small as β.

Theorem 1: Assume that the dataset $\mathcal{D}$ follows α-expansion and β-separation. Then, the training error of the model θ trained on $\mathcal{D}$ is bounded by the inverse of the total reduced neighborhood confidence $\sum_{\boldsymbol{x} \in \mathcal{D}} C_{\mathcal{N}}(\boldsymbol{x}; \mathcal{D})$ such that

$$\operatorname{Err}(\theta) \leq \frac{2 \cdot |\mathcal{D}| \cdot \operatorname{Err}(\theta_{\mathcal{M}})}{\sum_{\boldsymbol{x} \in \mathcal{D}} C_{\mathcal{N}}(\boldsymbol{x}; \mathcal{D})} + \frac{2 \cdot \alpha}{\alpha - 1} \cdot \beta \tag{4}$$

where $\theta_{\mathcal{M}}$ is a model trained with the supervised loss. Since β is usually very small, its effect on the error bound is negligible. Then, the bound highly depends on the total reduced neighborhood confidence. That is, as the total reduced neighborhood confidence increases, the error bound becomes tighter. This theorem provides provable noise robustness by amplifying authentic semantic correlations while suppressing label noise isolated from the intrinsic manifold.

Specifically, for each training sample, a corresponding set of representations $\boldsymbol{Z}^{(w)} = \{\boldsymbol{z}_1^{(w)}, \boldsymbol{z}_2^{(w)}, \ldots, \boldsymbol{z}_n^{(w)}\}$ is derived through weak augmentations. To quantify geometric proximity between samples $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ on the data manifold, we compute similarity using cosine distance

$$S_{ij} = \frac{\boldsymbol{z}_i^{(w)} \cdot \boldsymbol{z}_j^{(w)}}{\|\boldsymbol{z}_i^{(w)}\| \|\boldsymbol{z}_j^{(w)}\|}. \tag{5}$$

Subsequently, we construct a global manifold affinity graph $\boldsymbol{G} = (\boldsymbol{V}, \boldsymbol{E})$, where $\boldsymbol{V}$ represents the sample set and $\boldsymbol{E}$ is the edge set. The affinity matrix of $\boldsymbol{G}$ is defined by the representational similarity matrix $\boldsymbol{S} \in \mathbb{R}^{n \times n}$, where S_{ij} represents the representation similarity between samples $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$. Furthermore, $\forall i \in \{1, \ldots, n\}, S_{ii} = 0$ to prevent self-similarity from influencing the calculations.

Assume $\mathcal{B}$ is the selected mini-batch of samples, represented as $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_b\}$ with global indices $\{g_1, g_2, \ldots, g_b\}$. For each sample $\boldsymbol{x}_i$, we first identify the indices of the k-nearest manifold neighbors, denoted as $\{\mathcal{I}_{g_i,1}, \mathcal{I}_{g_i,2} \ldots, \mathcal{I}_{g_i,k}\}$. Then, we construct the similarity matrix $S_{\mathcal{B}}^k$ and the corresponding index matrix $\mathcal{N}_{\mathcal{B}}^k$

$$\boldsymbol{S}_{\mathcal{B}}^k = \begin{bmatrix} S\left[g_1, \mathcal{I}_{g_1,1}\right] & \cdots & S\left[g_1, \mathcal{I}_{g_1,k}\right] \\ \vdots & \ddots & \vdots \\ S\left[g_b, \mathcal{I}_{g_b,1}\right] & \cdots & S\left[g_b, \mathcal{I}_{g_b,k}\right] \end{bmatrix}$$

$$\boldsymbol{\mathcal{N}}_{\mathcal{B}}^k = \begin{bmatrix} \mathcal{I}_{g_1,1} & \cdots & \mathcal{I}_{g_1,k} \\ \vdots & \ddots & \vdots \\ \mathcal{I}_{g_b,1} & \cdots & \mathcal{I}_{g_b,k} \end{bmatrix}. \tag{6}$$

Notably, the global manifold affinity graph $\boldsymbol{G}$ is constructed at the start of each training epoch by calculating the pairwise similarities between all training samples. During each mini-batch, based on $\boldsymbol{G}$, we select the k-nearest neighbors of each sample to capture the local manifold structure, along with their corresponding global indices. Then, we construct the similarity matrix $S_{\mathcal{B}}^k$ and the corresponding index matrix $\mathcal{N}_{\mathcal{B}}^k$.

Then, we compute the MNC by aggregating neighborhood predictions weighted by the similarities of the manifold

$$C_{\mathcal{N}}(\boldsymbol{x}_i) = \frac{\sum_{j=1}^k S_{\mathcal{B}}^k[i,j]\, p\left(\boldsymbol{x}_{(\mathcal{N}_{\mathcal{B}}^k[i,j])}\right)}{\sum_{j=1}^k S_{\mathcal{B}}^k[i,j]}. \tag{7}$$

Ultimately, the manifold-aligned consistency loss enforces prediction consistency through KL divergence constraints

$$\mathcal{L}_{\text{mac}} = \sum_{i=1}^{b} \operatorname{KL}\left(p(\boldsymbol{x}_i) \| C_{\mathcal{N}}(\boldsymbol{x}_i)\right). \tag{8}$$

The proposed regularization embodies the core principle of MACCL: enforcing manifold-aligned consistency between the spectral–spatial representation space and the label space. Label distributions are aligned to locally smooth neighborhoods in

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the hyperspectral data through KL divergence constraints, and isolated noisy labels are corrected by propagating geometrically consistent signals across both spectral and spatial dimensions. This mechanism inherently avoids confirmation bias by anchoring label predictions to the manifold's geometric continuity, ensuring that decision boundaries strictly conform to the intrinsic structure of the hyperspectral data.

D. Noise-Tolerant Contrastive Representation Learning

In Section III-C, we mitigate the interference of isolated noisy labels by imposing local consistency constraints in the representation space, thereby enhancing the noise robustness in the label space. However, this requires our representation space to exhibit strong discriminative ability and enhanced geometry, ensuring that the data manifold accurately reflects its true categories. Existing CL methods are often disrupted by noisy labels, leading to unclear interclass structures within the representation space.

To address the issue of erroneous pair matching induced by noisy labels, we propose a confidence-guided pair selection strategy based on probabilistic affinity metrics. Specifically, confidence scores are computed through the inner products of normalized prediction probability vectors, followed by threshold filtering to retain high-confidence positive pairs. This design leverages the intrinsic property of deep networks to prioritize learning from high-confidence clean samples during the early stages of training, effectively suppressing error propagation from mislabeled instances. This is formalized as

$$\mathcal{L}'_{\text{ctr}}\left(\boldsymbol{x}_i,\boldsymbol{x}_j\right) = -\left(\langle\tilde{\boldsymbol{q}}_i,\tilde{\boldsymbol{z}}_j\rangle + \langle\tilde{\boldsymbol{q}}_j,\tilde{\boldsymbol{z}}_i\rangle\right)\mathbb{1}\left\{p\left(\boldsymbol{x}_i\right)^{\top} p\left(\boldsymbol{x}_j\right) \geq \tau\right\} \tag{9}$$

where $\tilde{\boldsymbol{q}}_k = \frac{\boldsymbol{q}_k}{\|\boldsymbol{q}_k\|_2}$ and $\tilde{\boldsymbol{z}}_k = \frac{\boldsymbol{z}_k}{\|\boldsymbol{z}_k\|_2}$. Following Simsiam [24], we define $\boldsymbol{q} = j(h(f(\boldsymbol{x})))$, $\boldsymbol{z} = \text{stopgrad}(h(f(\boldsymbol{x})))$, $h(f(\boldsymbol{x}))$ is an encoder network consisting of a backbone network and a projection head, and $j(\theta)$ is a prediction MLP. The classification head produces the probabilistic output $\boldsymbol{p}$, and τ is a confidence threshold.

We first compute the gradient information from (9), revealing how noisy labels distort the gradient and thereby disrupt the representation learning process. Specifically, we only consider the case where $\mathbb{1}\{p(\boldsymbol{x}_i)^{\top}p(\boldsymbol{x}_j) \geq \tau\} = 1$ because the loss term $\mathcal{L}'_{\text{ctr}}(\boldsymbol{x}_i,\boldsymbol{x}_j)$ is not calculated when $\mathbb{1}\{p(\boldsymbol{x}_i)^{\top}p(\boldsymbol{x}_j) \geq \tau\} = 0$. For simplicity, we assume that $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ are from the same class. Then, we can get

$$\mathcal{L}'_{\text{ctr}}\left(\boldsymbol{x}_i,\boldsymbol{x}_j\right) = -\left(\frac{\boldsymbol{q}_i}{\|\boldsymbol{q}_i\|_2}\cdot\frac{\boldsymbol{z}_j}{\|\boldsymbol{z}_j\|_2} + \frac{\boldsymbol{q}_j}{\|\boldsymbol{q}_j\|_2}\cdot\frac{\boldsymbol{z}_i}{\|\boldsymbol{z}_i\|_2}\right). \tag{10}$$

Let $a = \|\boldsymbol{q}_i\|_2$, $\boldsymbol{b} = \boldsymbol{q}_i$, $\boldsymbol{x} = \frac{\boldsymbol{z}_j}{\|\boldsymbol{z}_j\|_2}$, and $\boldsymbol{c} = \frac{\boldsymbol{b}}{a}$. Differentiating both sides of the equation $a^2 = \boldsymbol{b}^{\top}\boldsymbol{b}$, we obtain

$$2\ a\ \mathrm{d}a = 2\boldsymbol{b}^{\top}\mathrm{d}\boldsymbol{b}. \tag{11}$$

In the meanwhile

$$\begin{aligned}\partial\left(\frac{\boldsymbol{b}^{\top}\boldsymbol{x}}{a}\right) &= \frac{\mathrm{d}\left(\boldsymbol{b}^{\top}\boldsymbol{x}\right)a - \mathrm{d}a\boldsymbol{b}^{\top}\boldsymbol{x}}{a^2}\\ &\overset{(11)}{=} \frac{a\boldsymbol{x}^{\top}\mathrm{d}\boldsymbol{b}}{a^2} - \frac{\boldsymbol{b}^{\top}\mathrm{d}\boldsymbol{b}\boldsymbol{b}^{\top}\boldsymbol{x}}{a^3}\\ &= \frac{1}{a}\left(\boldsymbol{x}^{\top} - \boldsymbol{c}^{\top}\boldsymbol{x}\boldsymbol{c}^{\top}\right)\mathrm{d}\boldsymbol{b}.\end{aligned} \tag{12}$$

Substituting a, $\boldsymbol{b}$, $\boldsymbol{c}$, and $\boldsymbol{x}$ back into the equation, we obtain

$$\frac{\partial\mathcal{L}'_{\text{ctr}}\left(\boldsymbol{x}_i,\boldsymbol{x}_j\right)}{\partial\boldsymbol{q}_i} = -\frac{1}{\|\boldsymbol{q}_i\|_2}\left(\frac{\boldsymbol{q}_j}{\|\boldsymbol{q}_j\|_2} - \left(\frac{\boldsymbol{q}_i^{\top}\boldsymbol{q}_j}{\|\boldsymbol{q}_i\|_2\|\boldsymbol{q}_j\|_2}\right)\frac{\boldsymbol{q}_i}{\|\boldsymbol{q}_i\|_2}\right). \tag{13}$$

Since $\boldsymbol{z} = \text{stopgrad}(\boldsymbol{q}_i)$ based on the identity map $j(\theta)$, we define $c_i = 1/\|\boldsymbol{q}_i\|_2^2$, and thus we have

$$\left\|\frac{\partial\mathcal{L}'_{\text{ctr}}\left(\boldsymbol{x}_i,\boldsymbol{x}_j\right)}{\partial\boldsymbol{q}_i}\right\|_2^2 = c_i\left(1 - \left(\tilde{\boldsymbol{q}}_i^{\top}\tilde{\boldsymbol{q}}_j\right)^2\right). \tag{14}$$

Specifically, given two clean samples $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$, where $\boldsymbol{y}_i = \boldsymbol{y}_j$ and a mislabeled sample $\boldsymbol{x}_m$ with $\hat{\boldsymbol{y}}_m = \boldsymbol{y}_i$, in the early stages, the representation aligns such that $\boldsymbol{q}_i^T\boldsymbol{q}_j \to 1$ and $\boldsymbol{q}_i^T\boldsymbol{q}_m \approx 0$. However, as training progresses, deep networks begin to overfit to mislabeled data. At this point, incorrect contrastive pairs, such as $(\boldsymbol{x}_i,\boldsymbol{x}_m)$ and $(\boldsymbol{x}_j,\boldsymbol{x}_m)$, are erroneously pulled together, thereby leading to a gradient dominated by noisy pairs, as observed in (14), which ultimately governs the entire representation learning process

$$\begin{aligned}\left\|\frac{\partial\mathcal{L}'_{\text{ctr}}\left(\boldsymbol{x}_i,\boldsymbol{x}_m\right)}{\partial\boldsymbol{q}_i}\right\|_2^2 &= c_i\left(\underbrace{1 - \left(\tilde{\boldsymbol{q}}_i^{\top}\tilde{\boldsymbol{q}}_m\right)^2}_{\approx 1}\right)\\ \gg c_i\left(\underbrace{1 - \left(\tilde{\boldsymbol{q}}_i^{\top}\tilde{\boldsymbol{q}}_j\right)^2}_{\approx 0}\right) &= \left\|\frac{\partial\mathcal{L}'_{\text{ctr}}\left(\boldsymbol{x}_i,\boldsymbol{x}_j\right)}{\partial\boldsymbol{q}_i}\right\|_2^2.\end{aligned} \tag{15}$$

To address this issue, we propose a gradient regularization that leverages reliable pairs to guide spectral–spatial representations and mitigate incorrect contrastive pairs

$$\begin{aligned}&\mathcal{L}_{\text{ctr}}\left(\boldsymbol{x}_i,\boldsymbol{x}_j\right)\\ &= \left(\log\left(1 - \langle\tilde{\boldsymbol{q}}_i,\tilde{\boldsymbol{z}}_j\rangle\right) + \log\left(1 - \langle\tilde{\boldsymbol{q}}_j,\tilde{\boldsymbol{z}}_i\rangle\right)\right)\mathbb{1}\left\{p\left(\boldsymbol{x}_i\right)^{\top} p\left(\boldsymbol{x}_j\right) \geq \tau\right\}.\end{aligned} \tag{16}$$

Equation (16) still aims to learn similar representations for positive pairs. Specifically, minimizing (16), which is $(1 - \langle\tilde{\boldsymbol{q}}_i,\tilde{\boldsymbol{z}}_j\rangle)$, is mathematically equivalent to maximizing the similarity $\langle\tilde{\boldsymbol{q}}_i,\tilde{\boldsymbol{z}}_j\rangle$. Thus, similar to (9), the goal is to maximize the positive similarity, and its gradient is derived as follows:

$$\begin{aligned}\left\|\frac{\partial\mathcal{L}_{\text{ctr}}\left(\boldsymbol{x}_i,\boldsymbol{x}_j\right)}{\partial\boldsymbol{q}_i}\right\|_2^2 &= \frac{1}{1 - \tilde{\boldsymbol{q}}_i^{\top}\tilde{\boldsymbol{q}}_j}\left\|\frac{\partial\mathcal{L}'_{\text{ctr}}\left(\boldsymbol{x}_i,\boldsymbol{x}_j\right)}{\partial\boldsymbol{q}_i}\right\|_2^2\\ &= c_i\left(1 + \tilde{\boldsymbol{q}}_i^{\top}\tilde{\boldsymbol{q}}_j\right).\end{aligned} \tag{17}$$

Notably, the gradient from the correct pair $(\boldsymbol{x}_i,\boldsymbol{x}_j)$ is larger than from the noisy pair $(\boldsymbol{x}_i,\boldsymbol{x}_m)$

$$\begin{aligned}\left\|\frac{\partial\mathcal{L}_{\text{ctr}}\left(\boldsymbol{x}_i,\boldsymbol{x}_j\right)}{\partial\boldsymbol{q}_i}\right\|_2^2 &= c_i\left(\underbrace{1 + \tilde{\boldsymbol{q}}_i^{\top}\tilde{\boldsymbol{q}}_j}_{\approx 2}\right)\\ > c_i\left(\underbrace{1 + \tilde{\boldsymbol{q}}_i^{\top}\tilde{\boldsymbol{q}}_m}_{\approx 1}\right) &= \left\|\frac{\partial\mathcal{L}_{\text{ctr}}\left(\boldsymbol{x}_i,\boldsymbol{x}_m\right)}{\partial\boldsymbol{q}_i}\right\|_2^2.\end{aligned} \tag{18}$$

Compared with the gradients of the original loss in (15), the gradient in (18) from correct pairs is larger than from

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TABLE I
COMPREHENSIVE OVERVIEW OF THREE DATASETS EMPLOYED IN EXPERIMENTAL ANALYSES

NO.	KSC				HC				UP			
	Class Name	Train	Test	Total	Class Name	Train	Test	Total	Class Name	Train	Test	Total
1	Scrub	40	721	761	Strawberry	100	44635	44735	Asphalt	80	6551	6631
2	Willow swamp	40	203	243	Cowpea	100	22653	22753	Meadows	80	18569	18649
3	CP hammock	40	216	256	Soybean	100	10187	10287	Gravel	80	2019	2099
4	Slash pine	40	212	252	Sorghum	100	5253	5353	Trees	80	2984	3064
5	Oak/Broadleaf	40	121	161	Water spinach	100	1100	1200	Metal sheets	80	1265	1345
6	Hardwood	40	189	229	Watermelon	100	4433	4533	Bare Soil	80	4949	5029
7	Swap	40	65	105	Greens	100	5803	5903	Bitumen	80	1250	1330
8	Graminoid marsh	40	391	431	Trees	100	17878	17978	Bricks	80	3602	3682
9	Spartina marsh	40	480	520	Grass	100	9369	9469	Shadows	80	867	947
10	Cattail marsh	40	364	404	Red roof	100	10416	10516				
11	Salt marsh	40	379	419	Gray roof	100	16811	16911				
12	Mud flat	40	463	503	Plastic	100	3579	3679				
13	Water	40	887	927	Bare soil	100	9016	9116				
14					Road	100	18460	18560				
15					Bright object	100	1036	1136				
16					Water	100	75301	75401				
Total		640	4571	5211		1600	255930	257530		720	42056	42776

noisy pairs, enabling our regularization function to effectively prevent representation learning from being dominated by noisy pairs and avoid overfitting to corrupted labels.

Overall, NTCRL enhances model robustness to noisy labels by guiding the feature space with reliable pairs and improving the geometric structure of spectral and spatial features, thereby learning a noise-tolerant intrinsic manifold structure and preventing the model from learning incorrect representations.

E. MixUp-Based Supervised Learning

In supervised classification, the CELoss is typically used. In our framework, this metric is applied to assess the divergence between the predicted probabilities $p(\boldsymbol{x}_i)$ and their respective noisy labels $\hat{\boldsymbol{y}}_i$

$$\mathcal{L}_{ce}\left(\hat{\boldsymbol{y}}_i, p\left(\boldsymbol{x}_i\right)\right) = -\sum_{i=1}^{b} \hat{\boldsymbol{y}}_i \log\left(p\left(\boldsymbol{x}_i\right)\right). \tag{19}$$

Leveraging the effectiveness of the MixUp [50] technique in combating noisy labels, we employed it to enhance the model's robustness. Specifically, utilizing $\lambda \sim \text{Beta}(\alpha, \alpha)$, we combine $(\boldsymbol{x}_i, \hat{\boldsymbol{y}}_i)$ with a randomly selected $(\boldsymbol{x}_j, \hat{\boldsymbol{y}}_j)$ from the same mini-batch to generate $(\boldsymbol{x}_{\mathrm{i}}^{(m)}, \hat{\boldsymbol{y}}_{\mathrm{i}}^{(m)})$, effectively decelerating model convergence toward noisy labels

$$\begin{cases} \boldsymbol{x}_{\mathrm{i}}^{(m)} = \lambda \boldsymbol{x}_i + (1-\lambda)\boldsymbol{x}_j, \\ \hat{\boldsymbol{y}}_{\mathrm{i}}^{(m)} = \lambda \hat{\boldsymbol{y}}_i + (1-\lambda)\hat{\boldsymbol{y}}_j. \end{cases} \tag{20}$$

After MixUp processing, the form of CELoss changes to

$$\mathcal{L}_{\text{sup}} = -\sum_{i=1}^{b} \left(\lambda \hat{\boldsymbol{y}}_{\mathrm{i}}^{(m)} \log\left(p\left(\boldsymbol{x}_i\right)\right) + (1-\lambda)\hat{\boldsymbol{y}}_{\mathrm{i}}^{(m)} \log\left(p\left(\boldsymbol{x}_j\right)\right)\right). \tag{21}$$

The optimization of the framework is carried out under an integrated objective function that takes into account the representation contrastive loss, manifold-aligned consistency loss, and supervised learning loss

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \lambda_1 \mathcal{L}_{\text{ctr}} + \lambda_2 \mathcal{L}_{\text{mac}} \tag{22}$$

the loss weights, denoted as λ_1 and λ_2, are assigned values of $\lambda_1 = 1$ and $\lambda_2 = 10$, respectively. Notably, robust spectral–spatial representations are learned via NTCRL. In addition, the manifold-aligned consistency loss enhances prediction robustness by leveraging the geometric structure of neighboring samples, effectively mitigating model confirmation bias.

IV. EXPERIMENTAL RESULT

To comprehensively evaluate the effectiveness and generalizability of the proposed algorithm, we conducted extensive experiments on three publicly benchmark datasets, including comparative analysis with state-of-the-art methods, sensitivity analysis of algorithm parameters, and ablation studies.

A. Dataset Description

We conducted extensive experiments to evaluate the classification performance of the proposed MACCL on three public hyperspectral datasets, namely, KSC, HC, and UP. The specific details of the three datasets are summarized in Table I. To assess the model's robustness, we introduced varying levels of artificial noisy labels into the datasets, simulated real-world scenarios, and verified its resilience to label noise.

1) *Kennedy Space Center*[1] *(KSC):* The KSC dataset, derived from AVIRIS imagery over Kennedy Space Center in 1996, has a resolution of 512 × 614 pixels and maintains 176 bands after removal of noninformative bands due to water absorption and noise. It captures a spectral window from 400 to 2500 nm, categorizing 5211 pixels into 13 unique land cover types.
2) *WHU-Hi-HanChuan(HC)*[2]*:* The HC [51] dataset, acquired with a Headwall Nano-Hyperspec imaging sensor on June 17, 2016, in Hanchuan, Hubei, China, consists of a 1217 × 303 pixels image and 274

[1]http://www.ehu.eus/ccwintco/index.php/HyperspectralRemoteSensingScenes
[2]http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

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bands from 400 to 1000 nm. The study area contains 16 classes, which is a rural–urban fringe zone with buildings, water, and cultivated land.

3) *University of Pavia*[1] *(UP):* The UP dataset from a ROSIS sensor flight over Pavia, Italy, features a 610 × 340 HSI with 103 spectral bands ranging from 430 to 860 nm and a resolution of 1.3 m/pixel. After refining to remove noise, the dataset identifies nine land cover types across 42 776 labeled pixels, primarily focusing on crops in various growth stages.

B. Data Augmentations

Existing research has demonstrated that the performance of CL heavily depends on the selection of appropriate data augmentation strategies [54]. In light of this, we employ two data augmentation approaches: weak augmentation and strong augmentation.

Weak Augmentation: This primarily utilizes simple spatial rotation. Its implementation involves randomly rotating each training pixel patch by 0°, 90°, 180°, or 270° along the spatial dimensions. This transformation introduces subtle variations in spatial orientation while preserving spectral integrity, helping the model focus on fine spatial changes and improving sensitivity to spatial–spectral features.

Strong Augmentation: This primarily utilizes random spectral shift [55]. Its implementation involves randomly selecting a portion of the spectral bands and shifting half of the spectral values along the spectral dimension horizontally, while the other half is shifted vertically. This transformation significantly alters the spectral information, enhancing the model's robustness by exploiting hyperspectral data diversity.

In practice, weak augmentation is utilized for training the classifier, and the resulting low-dimensional representations obtained from the projection head are used to compute nearest-neighbor relations, while strong augmentation is applied to instance-level CL. Formally, Given an input patch $\boldsymbol{x} \in \mathbb{R}^{L\times L\times B}$, we generate three augmented versions: $\boldsymbol{x}^{(w)}$ through a weak transformation and $\boldsymbol{x}^{(s_1)}$ and $\boldsymbol{x}^{(s_2)}$ via two independent strong transformations.

C. Experimental Setup

This study evaluates the MACCL model on three benchmark HSI datasets, namely, KSC, HC, and UP. The experiments are conducted using the PyTorch 2.0.1 framework, utilizing hardware resources including a GeForce RTX 3090, 24 GB of memory, and CUDA 12.40. Within the MACCL framework, the Adam optimizer is employed, with a learning rate set to 0.0001 and a batch size of 64. The backbone of the MACCL is based on the state-of-the-art TCRL framework, which is employed for HIC with noisy labels [22].

The noisy labels are synthetically generated by manually corrupting the true labels with a noise ratio $r \in (0, 1)$. For each class, a random subset of r proportion of the training instances is selected, and their labels are changed to one of the other $c-1$ classes with a probability of $r/(c-1)$. This means that the probability of a label being flipped to any other class is the same. The noise ratio r controls the proportion of noisy labels, simulating real-world labeling errors.

To quantitatively assess the performance of the proposed model, three evaluation metrics are considered: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (Kappa). To ensure the reliability of the results, ten repetitions of random experiments are performed, and the average values and standard deviations are calculated. During the inference stage, the final trained model is utilized for prediction.

D. Comparison With the State-of-the-Art Methods

To evaluate the effectiveness of the MACCL framework, we compare it with several advanced HIC methods for handling noisy labels, including DPNLD [52], MSSG [5], DCRN [15], TCRL [22], MASS [53], SPTNet-NTLA [4], and GLHLE [7]. Among these methodologies, DPNLD is a data-driven approach, utilizing sample selection mechanisms to remove or correct compromised labels to enhance the quality of the training dataset. Conversely, MSSG, DCRN, TCRL, GLHLE, and SPTNet-NTLA employ model-driven strategies, developing frameworks for supervised learning that demonstrate tolerance to label noise. MASS is an advanced HIC method.

To ensure the rigor and fairness of our experimental comparisons, this study strictly adheres to the parameter settings recommended by the original publications and ensures that the experimental design conforms to the protocols detailed in Section IV-C. All comparative methods are implemented by us using the released code. In order to thoroughly investigate MACCL and the comparative methods, we followed the guidelines provided in [21] and recorded the results for each dataset as the noise rate r varied from 0.1 to 0.4.

1) Experiments on the KSC Dataset: Table II lists the classification performance of various methods on the KSC dataset. From the experimental results, we have the following findings. First, the research demonstrates that at lower noise levels r, each method maintains high classification accuracy. However, as r escalates, a distinct pattern of performance deterioration becomes evident. Notably, MASS experiences a notable reduction in performance, as r increases from 0.1 to 0.4. Since its algorithm is not noise-resistant, it is easily disturbed by noisy labels. In this scenario, MACCL shows minimal performance decline, maintaining an OA above 99%, demonstrating strong robustness against label noise.

Second, SPTNet-NTLA, GLHLE, and TCRL leverage noise-tolerant learning algorithms and triplet contrastive loss functions, significantly boosting the models' resilience to noisy labels. Furthermore, MACCL is instrumental in mitigating overfitting to noisy labels by exploiting the inherent structure of the data in the feature space, maintaining optimal performance even with escalating noise levels.

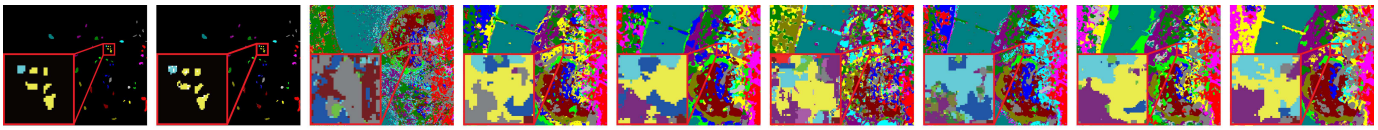
Finally, although DCRN and TCRL show commendable performance, the MACCL method consistently outperforms across different noise ratio conditions, achieving improvements over DCRN by 0.85% in OA, 0.98% in AA, and 0.96% in Kappa at $r = 0.1$, and 2.59%, 2.88%, and 2.87% at $r = 0.4$. Similarly, enhancements over TCRL are 0.37%, 0.51%, and 0.41% at $r = 0.1$ and 0.95%, 1.23%, and 1.04% at $r = 0.4$. Moreover, Fig. 4 demonstrates that, even with a

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TABLE II

COMPARATIVE ANALYSIS OF CLASSIFICATION PERFORMANCE FOR DPNLD, MSSG, DCRN, TCRL, MASS, SPTNET-NTLA, GLHLE, AND MACCL ON THE KSC DATASET ACROSS VARIOUS NOISE RATES

r	Evaluation index	DPNLD [52]	MSSG [5]	DCRN [15]	TCRL [22]	MASS [53]	SPTNet-NTLA [4]	GLHLE [7]	MACCL
0.1	OA(%)	89.76±1.12	84.53±1.73	98.84±1.57	99.32±0.26	93.45±0.46	98.58±1.14	99.48±0.23	**99.69±0.11**
	AA(%)	84.79±1.04	77.70±2.74	98.40±2.25	98.87±0.47	91.63±0.29	97.67±1.64	99.18±0.35	**99.38±0.40**
	Kappa(%)	88.56±1.24	82.64±1.94	98.69±1.76	99.24±0.30	92.66±0.52	98.41±1.27	99.42±0.12	**99.65±0.12**
0.2	OA(%)	88.75±1.72	84.11±1.75	98.52±1.06	99.05±0.55	90.50±0.96	98.01±1.59	99.26±0.89	**99.61±0.10**
	AA(%)	83.89±1.89	77.46±2.27	97.75±1.75	98.50±0.76	87.59±2.03	97.48±2.80	98.83±0.55	**99.30±0.16**
	Kappa(%)	87.44±1.91	82.17±1.96	98.34±1.19	98.93±0.62	89.35±1.08	98.04±1.78	99.17±0.11	**99.57±0.10**
0.3	OA(%)	85.94±1.55	83.97±1.94	98.11±0.49	98.77±0.56	87.17±1.67	97.22±1.97	98.94±0.77	**99.32±0.18**
	AA(%)	80.95±1.49	77.26±2.75	97.19±0.94	98.01±0.90	85.65±1.99	95.99±2.96	98.23±0.37	**98.68±0.62**
	Kappa(%)	84.31±1.72	82.02±2.16	97.87±0.55	98.62±0.63	85.66±1.85	96.88±2.20	98.81±0.98	**99.25±0.20**
0.4	OA(%)	83.53±1.89	83.59±1.97	96.47±1.08	98.11±0.97	79.54±5.00	96.53±1.47	98.79±0.74	**99.06±0.20**
	AA(%)	78.27±1.82	77.07±2.71	95.39±1.49	97.04±1.32	76.79±4.30	94.51±2.75	97.64±0.37	**98.27±0.41**
	Kappa(%)	81.64±2.09	81.58±2.22	96.06±1.19	97.89±1.08	77.20±5.43	95.99±1.65	98.65±0.94	**98.93±0.22**



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 4. Classification maps generated using various approaches on the KSC dataset with a noise rate of r = 0.4. (a) Ground-truth map. (b) Noisy label map (white points indicate noisy pixels). (c) MSSG. (d) DCRN. (e) TCLR. (f) MASS. (g) SPTNet-NTLA. (h) GLHLE. (i) Proposed MACCL.

TABLE III

COMPARATIVE ANALYSIS OF CLASSIFICATION PERFORMANCE FOR DPNLD, MSSG, DCRN, TCRL, MASS, SPTNET-NTLA, GLHLE, AND MACCL ON THE HC DATASET ACROSS VARIOUS NOISE RATES

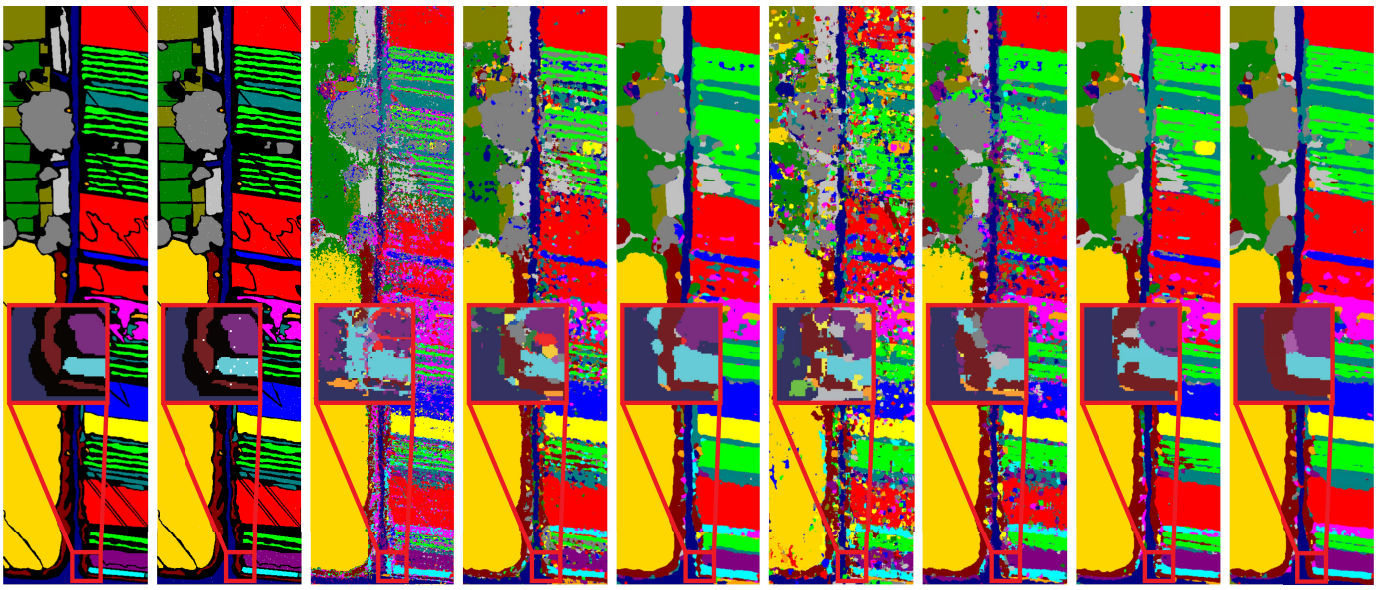
r	Evaluation index	DPNLD [52]	DCRN [15]	TCRL [22]	SSRN [56]	MASS [53]	SPTNet-NTLA [4]	GLHLE [7]	MACCL
0.1	OA(%)	77.64±1.21	80.38±0.16	93.67±0.63	95.75±0.14	86.72±1.79	94.61±0.06	95.32±0.05	**96.69±0.14**
	AA(%)	75.36±0.97	78.24±1.32	92.62±0.86	92.35±0.51	85.96±0.68	93.28±0.07	91.15±0.03	**94.40±0.72**
	Kappa(%)	74.24±1.36	77.31±0.22	93.75±1.08	95.04±0.19	84.54±2.02	93.71±0.08	94.52±0.05	**96.13±0.17**
0.2	OA(%)	76.98±0.45	80.16±0.36	92.24±1.16	94.88±0.02	85.31±1.74	92.85±0.11	93.45±0.54	**95.96±0.08**
	AA(%)	74.41±0.64	78.21±1.05	92.47±0.78	89.99±0.09	85.16±1.35	92.42±0.28	86.33±6.99	**92.52±0.21**
	Kappa(%)	73.46±0.49	77.07±0.40	90.96±1.56	94.02±0.03	82.98±1.95	91.65±0.16	92.36±0.72	**95.28±0.10**
0.3	OA(%)	76.28±0.84	79.77±0.60	90.12±3.61	94.41±0.06	80.28±1.52	90.90±1.37	92.56±0.05	**95.55±0.14**
	AA(%)	73.79±0.52	77.83±1.06	90.52±1.69	89.06±0.19	79.00±0.78	90.41±1.79	84.82±4.32	**90.03±0.46**
	Kappa(%)	72.66±0.89	76.63±0.65	88.52±4.75	93.47±0.08	77.13±1.72	89.40±1.85	91.31±0.07	**94.80±0.16**
0.4	OA(%)	74.07±1.18	78.98±0.89	85.83±3.27	93.59±0.35	68.19±3.26	85.87±1.28	91.38±0.05	**94.37±0.86**
	AA(%)	71.92±1.13	77.70±0.67	83.64±3.70	86.28±0.68	69.32±1.54	85.88±1.85	83.64±0.18	**89.25±0.31**
	Kappa(%)	70.16±1.31	75.75±0.95	83.52±3.74	92.53±0.47	63.74±3.35	83.60±1.72	89.94±0.07	**93.43±0.10**

noise rate of r = 0.4, the proposed MACCL yields smoother classification maps than alternatives, particularly in noisy label regions (white points). These improvements stem from MACCL's noise-tolerant CL, which selects high-confidence pairs to enhance discriminative representation.

2) Experiments on the HC Dataset: In the HC dataset, classification performances of various methodologies are summarized in Table III. Consistent with observations from the KSC dataset, an increase in the noise ratio r results in significant performance degradation for DPNLD and MASS, predominantly attributed to escalating confirmation bias within these models as the noise level increases. Among the methodologies demonstrating superior effectiveness, DCRN, TCRL, SPTNet-NTLA, GLHLE, and MACCL approaches stand out. Specifically, as the noise ratio extends to 0.4, the MACCL approach outperforms the second-best TCRL in key metrics of OA, AA, and Kappa by increments of 0.78%, 2.97%, and 0.90%, respectively. Moreover, Fig. 5 demonstrates that, even in scenarios with a noise rate of r = 0.4, the proposed MACCL consistently yields smoother classification maps compared with alternative approaches. This consistent performance across various noise levels is mainly due to MACCL's integrated design, which combines noise-tolerant CL with manifold-aligned consistency learning to mitigate the influence of noisy labels and prevent overfitting, ultimately ensuring reliable classification accuracy.

3) Experiments on the UP Dataset: On the UP dataset, the classification performances of various methodologies are enumerated in Table IV. Consistent with findings from the

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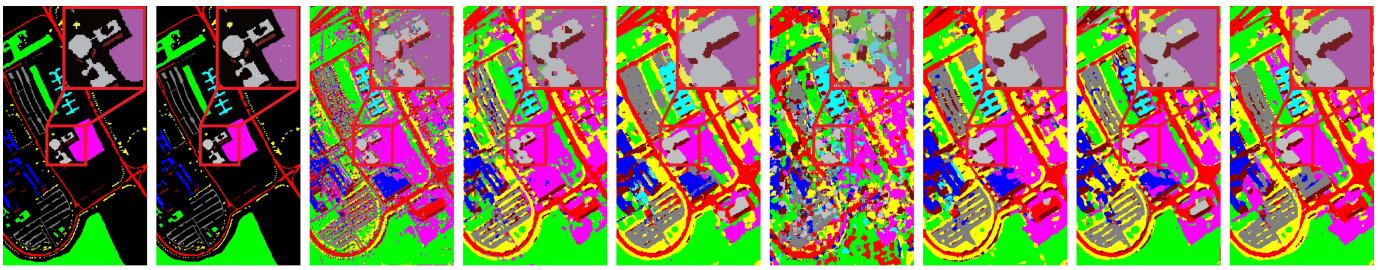
(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 5. Classification maps generated using various approaches on the HC dataset with a noise rate of r = 0.4. (a) Ground-truth map. (b) Noisy label map (white points indicate noisy pixels). (c) MSSG. (d) DCRN. (e) TCLR. (f) MASS. (g) SPTNet-NTLA. (h) GLHLE. (i) Proposed MACCL.

TABLE IV

COMPARATIVE ANALYSIS OF CLASSIFICATION PERFORMANCE FOR DPNLD, MSSG, DCRN, TCRL, MASS, SPTNET-NTLA, GLHLE, AND MACCL ON THE UP DATASET ACROSS VARIOUS NOISE RATES

r	Evaluation index	DPNLD [52]	MSSG [5]	DCRN [15]	TCRL [22]	MASS [53]	SPTNet-NTLA [4]	GLHLE [7]	MACCL
0.1	OA(%)	81.82±2.79	93.12±1.25	98.10±0.87	98.72±0.61	92.00±2.41	98.34±0.35	98.02±0.69	**99.11±0.21**
	AA(%)	78.37±2.36	94.21±0.65	98.38±0.45	98.12±0.87	91.24±1.15	98.33±0.13	96.31±0.36	**98.58±0.43**
	Kappa(%)	76.70±3.27	90.95±1.59	97.49±1.15	98.30±0.80	89.52±3.01	97.80±0.46	97.37±0.88	**98.78±0.28**
0.2	OA(%)	77.03±2.87	93.03±0.96	97.23±1.10	98.01±0.55	84.42±1.57	97.80±0.34	97.64±0.44	**98.46±0.32**
	AA(%)	74.09±2.08	94.14±0.48	97.03±0.74	97.05±1.07	85.36±1.70	97.22±0.27	95.74±0.70	**97.55±0.76**
	Kappa(%)	71.15±3.28	90.83±1.22	96.33±1.45	97.37±0.73	79.87±1.93	97.08±0.44	96.87±0.81	**97.89±0.43**
0.3	OA(%)	73.16±3.56	92.18±1.36	96.10±0.91	97.03±1.12	74.74±2.29	97.33±0.43	96.85±0.28	**97.74±0.86**
	AA(%)	70.40±3.84	93.59±0.54	96.45±0.88	95.74±1.52	75.82±3.06	96.67±0.44	94.99±0.26	**96.91±0.99**
	Kappa(%)	66.56±4.07	89.73±1.72	94.84±1.19	96.07±1.45	68.21±2.66	96.45±0.57	95.84±0.86	**96.68±0.94**
0.4	OA(%)	66.44±2.61	91.24±1.37	91.69±2.81	96.02±1.28	64.11±3.54	96.22±1.10	96.26±0.91	**96.82±0.93**
	AA(%)	69.70±2.64	92.52±1.19	93.80±1.87	94.74±1.86	67.05±1.20	95.43±0.70	93.49±0.09	**95.51±1.53**
	Kappa(%)	59.25±2.72	88.51±1.75	89.14±3.58	94.75±1.66	55.38±3.75	95.00±1.45	95.05±0.28	**95.66±0.87**



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 6. Classification maps generated using various approaches on the UP dataset with a noise rate of r = 0.4. (a) Ground-truth map. (b) Noisy label map (white points indicate noisy pixels). (c) MSSG. (d) DCRN. (e) TCLR. (f) MASS. (g) SPTNet-NTLA. (h) GLHLE. (i) Proposed MACCL.

KSC and HC datasets, as the noise ratio r increases, our MACCL method exhibits the most robust performance. For example, as the noise ratio r increases from 0.1 to 0.4, MACCL consistently outperforms the other models, and on the OA metric, it outperforms DPNLD by 17.29%, 21.43%, 24.58%, and 30.38%; TCRL by 0.39%, 0.45%, 0.71%, and 0.8%; and SPTNet-NTLA by 0.77%, 0.66%, 0.41%, and 0.6%. This data underscore the consistent and stable performance of the MACCL method regardless of the noise level. In addition, Fig. 6 demonstrates that, even in scenarios with a noise rate

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TABLE V
PER-CLASS ACCURACY UNDER VARYING NOISE RATES ON THE KSC DATASET

Class	$r = 0.1$	$r = 0.2$	$r = 0.3$	$r = 0.4$
1	99.96	99.93	99.98	99.92
2	99.72	99.72	99.87	99.13
3	99.19	98.10	96.84	96.02
4	98.34	97.06	95.43	95.08
5	97.20	97.64	95.02	92.26
6	99.32	99.54	99.43	99.11
7	99.38	98.34	97.60	97.16
8	99.76	99.77	99.85	99.51
9	99.97	99.95	99.93	99.65
10	100.00	100.00	100.00	99.79
11	99.91	99.96	99.53	99.38
12	99.84	99.82	99.70	99.42
13	100.00	100.00	100.00	100.00
OA(%)	99.69±0.11	99.61±0.10	99.32±0.18	99.06±0.20
AA(%)	99.38±0.40	99.30±0.16	98.68±0.62	98.27±0.41
Kappa(%)	99.65±0.12	99.57±0.10	99.25±0.20	98.93±0.22

of $r = 0.4$, the proposed MACCL consistently yields smoother classification maps compared with alternative approaches.

Overall, in the experimental analysis of the KSC, HC, and UP datasets, we found that the impact of noisy labels varies with dataset complexity. The simpler KSC dataset, with fewer classes and less complex geographical features, is less affected by noisy labels, resulting in a slower performance decline. In contrast, the more complex HC and UP datasets, where distinguishing between classes is more difficult, are more sensitive to noisy labels, leading to a greater performance drop, especially under high noise conditions. The smaller decline in KSC is mainly due to its simpler structure, where noisy labels have less impact.

E. Performance Under High Noise Rates

In this section, to evaluate the performance of different methods under high noise, we conducted comparative experiments at different levels of high noise ratios. Specifically, we set the noise ratios to 50%, 60%, and 70%. Fig. 7(a)–(i) shows the OA, AA, and Kappa values of different methods at these varying noise ratios. From Fig. 7, it can be seen that as the noise rate increases, the accuracy of all methods declines to varying degrees. However, our MACCL framework consistently maintains higher accuracy. Even when the noise rate reaches its maximum value, the proposed MACCL still achieves an OA of 90% on the KSC dataset, while the accuracy of other methods significantly decreases. This is primarily due to its innovative manifold alignment consistency mechanism and noise-tolerant CL strategy, which effectively suppress the interference of noisy labels. Meanwhile, the optimization of the feature space further enhances the model's discriminative ability, allowing MACCL to exhibit stronger adaptability and stability in high-noise environments.

F. t-SNE Visualization

In this section, as shown in Fig. 8, we visualize the learned features of the KSC, HC, and UP datasets at a 0.4 noise rate using t-SNE, directly comparing the advanced classifier MASS, the robust algorithm SPTNet-NTLA, and our proposed MACCL. For the very large HC dataset, only the first 20% of its samples are plotted. The experimental results reveal that MASS lacks robustness to label noise because its feature distribution is heavily disrupted, and class boundaries become obscured. In contrast, SPTNet-NTLA generates a more orderly distribution overall, yet still exhibits noticeable class overlap. Ultimately, MACCL produces the most favorable pattern, showing clearer interclass separation, tighter yet suitably diverse intraclass clusters, and larger margins between classes. These observations collectively demonstrate that MACCL can learn more discriminative, noise-resilient representations and establish more reliable decision boundaries under noisy labels.

TABLE VI
PER-CLASS ACCURACY UNDER VARYING NOISE RATES ON THE HC DATASET

Class	$r = 0.1$	$r = 0.2$	$r = 0.3$	$r = 0.4$
1	98.72	98.72	97.40	97.36
2	98.97	98.74	98.77	97.73
3	97.06	96.23	94.92	94.87
4	98.35	97.92	97.00	96.01
5	99.63	96.81	91.38	87.08
6	79.05	75.12	70.73	56.34
7	87.84	81.90	81.52	80.43
8	97.94	97.35	97.15	94.91
9	91.29	88.35	88.02	88.23
10	98.50	98.30	98.43	98.28
11	94.01	91.88	91.59	90.75
12	93.09	90.70	85.20	79.04
13	81.43	80.57	81.21	78.67
14	98.13	97.89	97.74	96.91
15	89.68	88.54	86.58	87.06
16	99.98	99.99	99.97	99.81
OA(%)	96.69±0.14	95.96±0.08	95.55±0.14	94.37±0.86
AA(%)	94.40±0.72	92.52±0.21	90.03±0.46	89.25±0.31
Kappa(%)	96.13±0.17	95.28±0.10	94.80±0.16	93.43±0.10

TABLE VII
PER-CLASS ACCURACY UNDER VARYING NOISE RATES ON THE UP DATASET

Class	$r = 0.1$	$r = 0.2$	$r = 0.3$	$r = 0.4$
1	99.92	99.67	99.40	98.97
2	99.81	99.52	99.51	98.62
3	97.02	96.28	95.99	94.44
4	97.73	97.83	96.32	96.23
5	99.50	99.16	98.29	98.12
6	98.68	97.02	94.13	91.49
7	98.71	97.87	97.10	93.10
8	96.82	95.12	94.12	94.19
9	98.38	97.61	96.90	93.74
OA(%)	99.11±0.21	98.46±0.32	97.74±0.86	96.82±0.93
AA(%)	98.58±0.43	97.55±0.76	96.91±0.99	95.51±1.53
Kappa(%)	98.78±0.28	97.89±0.43	96.68±0.94	95.66±0.87

G. Per-Class Accuracy Under Varying Noise Rates

In this section, to further validate the category robustness of our model, we computed the accuracy for each class in the KSC, HC, and UP datasets under varying noise rates (r = 0.1–0.4), with the results shown in Tables V–VII. Based on the experimental results, we made the following observations.

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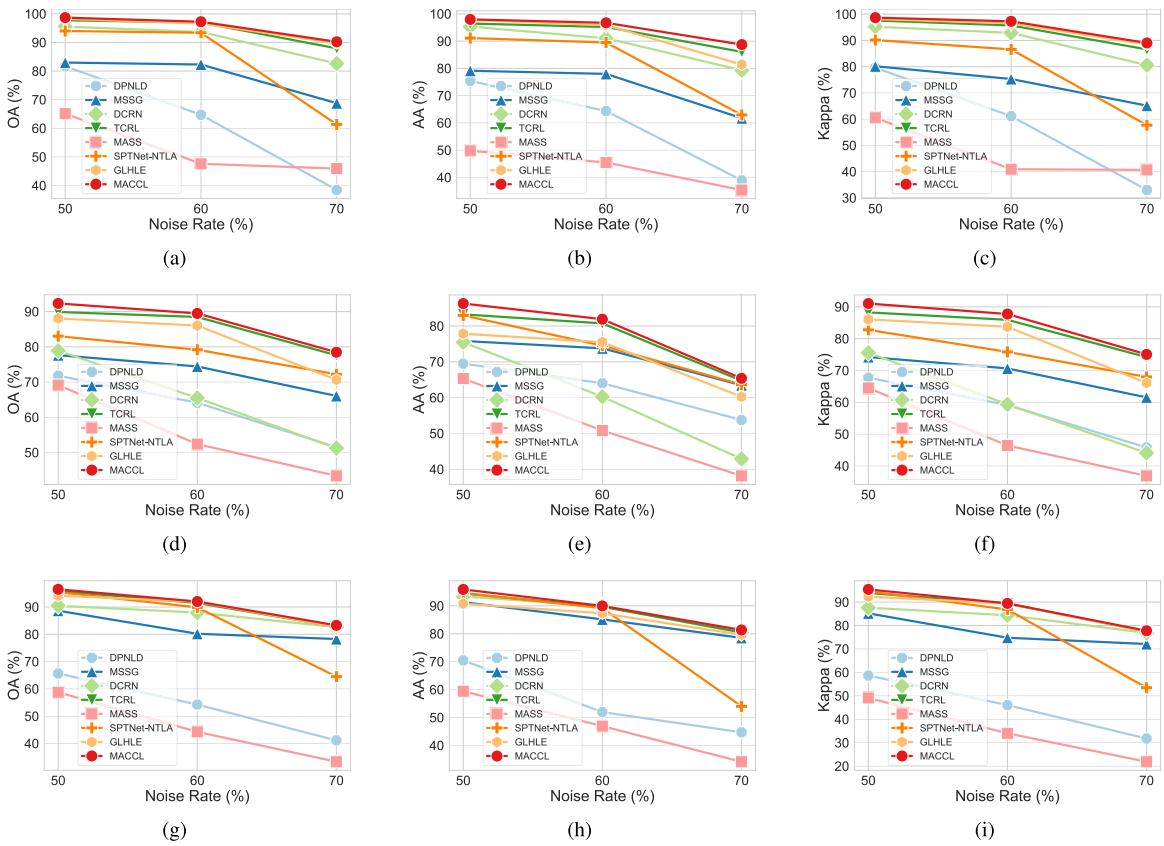


Fig. 7. Performance of different methods under varying levels of severe noise. (a)–(c) KSC. (d)–(f) HC. (g)–(i) UP.

First, as the noise rate increases, the accuracy of nearly all classes gradually decreases, mainly due to the interference of noisy labels with the decision boundaries, making it harder for the model to correctly classify. Second, in the UP dataset, the accuracy of class 6 dropped by about 7.19% (from 98.68% to 91.49%), while the accuracy of class 1 decreased by less than 1% (from 99.92% to 98.97%). This difference can be attributed to the more complex spectral features and higher intraclass variability of class 6, making it more susceptible to noise interference, while class 1, with simpler spectral features, is less affected by noise. In addition, we observed that some classes experienced slight improvements in performance as the noise rate increased. For example, in the KSC dataset, the accuracy of class 5 increased by 0.44% from a noise rate of 0.1 to 0.2. This may be due to the class having simpler and more stable spectral features, which are less affected by noise, and the increase in noise may prompt the model to focus more on these stable features, thereby improving performance. In summary, at a noise rate of 0.4, the worst-performing classes in the KSC and UP datasets still maintained high accuracies of 92.26% and 91.49%, respectively, demonstrating that our model exhibits strong robustness across most classes.

H. Parameter Sensitivity Analysis

In this section, parameter sensitivities are evaluated for the number of neighbors k, confidence threshold τ, and balance coefficients λ_1 and λ_2. Experiments are performed on the KSC, HC, and UP datasets, maintaining a noise ratio of 0.4 and utilizing OA as the performance metric.

1) Effect of Different Numbers of Neighbors k: We investigated the influence of neighbor count k across values 2, 6, 10, 20, 30, and 40, as shown in Fig. 9(a). According to manifold learning theory, the local neighborhood structure is crucial for capturing the intrinsic data geometry. As k increases, the model incorporates more local information, helping to reduce noisy labels. Initially, as k increases from 2 to 10, accuracy improves because more neighbors better reflect the data's local distribution, aligning with the continuity assumption, where geometrically close points likely share the same label.

However, beyond a certain threshold (around $k = 20$), accuracy declines. This is due to the over-smoothing effect in manifold learning, where too many neighbors make labels too uniform, diminishing local geometric distinctions. This excessive smoothing leads to a loss of fine-grained patterns, negatively impacting generalization. Therefore, based on both theory and experimental results, setting $k = 20$ offers a balanced tradeoff, preserving local structure while avoiding over-smoothing, ensuring stable performance.

2) Effect of Confidence Threshold τ: In our experiments, we examined the impact of setting the confidence threshold τ between 0.2 and 0.9, as shown in Fig. 9. From a manifold learning perspective, τ controls prediction consistency in the label space by ensuring that nearby points on the data manifold, which should have similar labels, correspond to positive pairs with similar feature distributions in CL.

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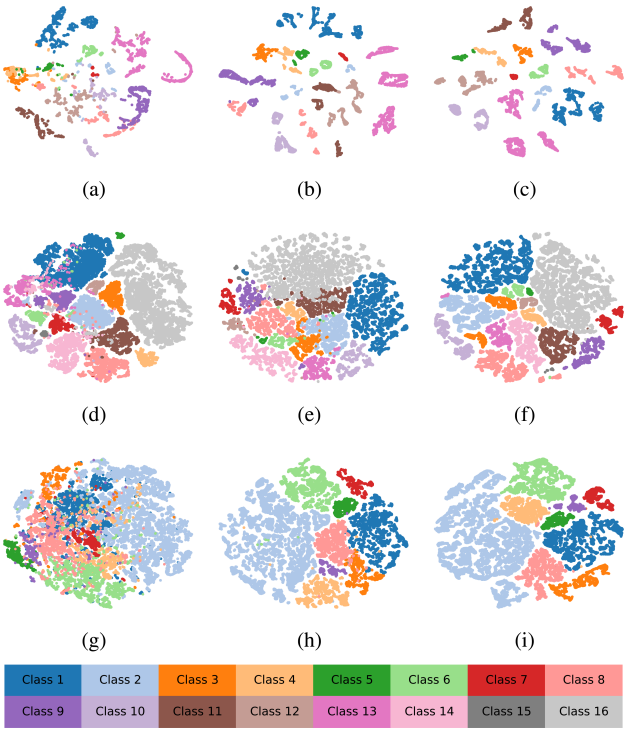


Fig. 8. t-SNE visualizations of feature embeddings on three benchmark datasets. (a)–(c) KSC. (d)–(f) HC. (g)–(i) UP. Within each dataset, the visualizations correspond to the methods: MASS, SPTNet-NTLA, and MACCL, respectively.

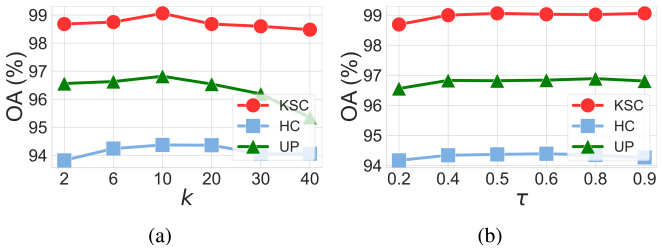


Fig. 9. Sensitivity analysis of various hyperparameters on MACCL is illustrated by OA curves computed for the KSC, HC, and UP datasets. (a) Number of neighbors k. (b) Confidence threshold τ.

A lower τ allows more pairs, including noisy ones, to be considered positive, while a higher τ ensures only clean pairs with stronger consistency are selected.

As shown in Fig. 9, when τ is set to 0.2, the model's performance drops significantly due to noisy pairs. Performance stabilizes beyond a certain threshold, with $\tau = 0.5$ achieving the best results. This balance allows the model to focus on high-confidence pairs and improve stability.

3) Effect of Joint Analysis of Regularization Weights λ_1 and λ_2: We conducted a joint analysis of the sensitivity of two regularization parameters, λ_1, λ_2, across three datasets. As described in Section III, λ_1 controls the weight of the noise-tolerant robust loss, while λ_2 regulates the strength of the manifold-aligned consistency loss. To determine the optimal parameter configuration, we explored λ_1 and λ_2 within the range $\{1e^{-2}, 1e^{-1}, 1e^{+0}, 1e^{+1}, 1e^{+2}\}$. Fig. 10 illustrates the OA

TABLE VIII

ABLATION STUDY RESULTS OF DIFFERENT COMPONENTS OF THE MACCL FRAMEWORK ON THE KSC, HC, AND UP DATASETS

Dataset	Method	OA	AA	Kappa
KSC	MACCL	**99.06±0.20**	**98.27±0.41**	**98.93±0.22**
	w/o $\mathcal{L}_{sup}$	98.79±0.22	97.69±0.96	98.64±0.25
	w/o $\mathcal{L}_{ctr}$	98.91±0.31	98.20±0.47	98.78±0.36
	w/o $\mathcal{L}_{mac}$	98.43±0.36	97.39±0.75	98.24±0.40
HC	MACCL	**94.37±0.86**	**89.25±0.31**	**93.43±0.10**
	w/o $\mathcal{L}_{sup}$	93.92±0.26	89.06±0.24	92.90±0.29
	w/o $\mathcal{L}_{ctr}$	93.34±0.53	86.04±1.65	92.23±0.62
	w/o $\mathcal{L}_{mac}$	93.01±0.29	85.50±0.81	91.83±0.33
UP	MACCL	**96.82±0.93**	**95.51±1.53**	**95.66±0.87**
	w/o $\mathcal{L}_{sup}$	96.28±0.78	95.36±0.49	95.08±0.31
	w/o $\mathcal{L}_{ctr}$	96.13±1.83	94.96±2.27	94.89±2.35
	w/o $\mathcal{L}_{mac}$	95.08±0.97	93.41±1.48	93.43±1.29

for different parameter combinations across the three datasets. The results show that when the values of λ_1 and λ_2 are either too small or too large, particularly on the UP and HC datasets, the model performance significantly deteriorates. Furthermore, the best combinations of λ_1 and λ_2 typically lie within the range $\{1e^{+0}, 1e^{+1}\}$. After a comprehensive evaluation of the experimental results across the three datasets, we selected $\lambda_1 = 1e^{+0}$ and $\lambda_2 = 1e^{+1}$ as the default parameter configuration for the model, as this combination ensures effective manifold alignment while maintaining noise robustness, leading to stable and superior performance.

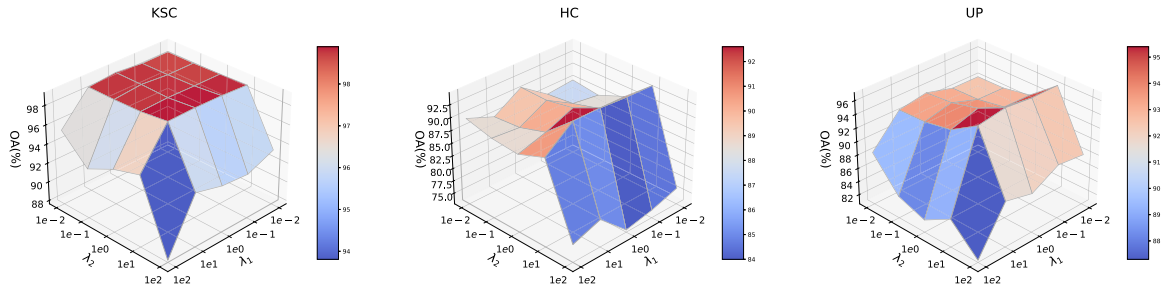
I. Ablation Experiments

In this ablation study, we evaluate the contributions of three key components in the context of HIC: the Mixup-based supervised loss, the noise-tolerant CL loss, and the manifold-aligned consistency loss, as detailed in Table VIII. Initially, the Mixup-based supervised loss smooths decision boundaries by linearly interpolating hyperspectral cubes and their corresponding labels, thereby implicitly regularizing the model and reducing sensitivity to sporadic noisy annotations. Next, the noise-tolerant CL loss employs a confidence-guided sample pair selection mechanism that focuses on high-confidence (clean) pairs and integrates a gradient optimization strategy where gradients from these reliable pairs significantly outweigh those from noisy pairs. This ensures that discriminative feature learning is predominantly driven by reliable spectral–spatial data. Finally, the manifold-aligned consistency loss aligns the predicted label distributions with the intrinsic spectral–spatial data manifold, effectively mitigating confirmation bias and reducing the detrimental impact of noisy labels. Collectively, these components synergize to enhance the model's robustness and generalization performance, ensuring reliable classification in the complex domain of hyperspectral imaging.

J. Robustness of Local Manifold Structures

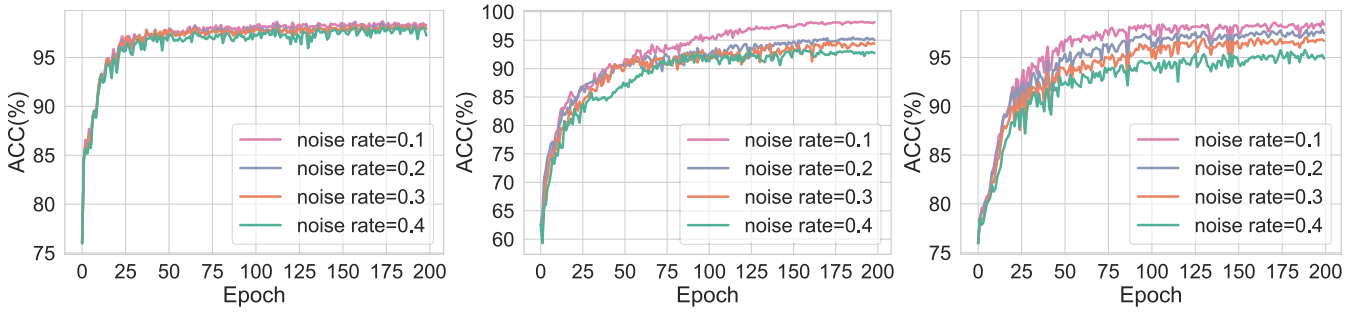
In this section, we analyze the consistency between the golden labels of each training sample and its local manifold

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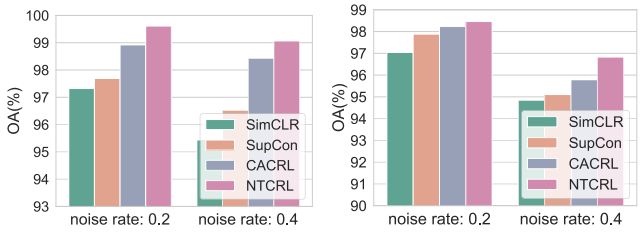
(a) (b) (c)

Fig. 10. Comparative analysis of hyperparameters λ_1 and λ_2 on the OA of MACCL. (a) KSC. (b) HC. (c) UP.



(a) (b) (c)

Fig. 11. Evolution of manifold neighborhood consistency across KSC, HC, and UP datasets under different noise rates. (a) KSC. (b) HC. (c) UP.



(a) (b)

Fig. 12. Comparison of NTCRL with existing CRL approaches on the HC and UP datasets with noise rates of 0.2 and 0.4. (a) KSC. (b) UP.

structure under different noise levels. High local manifold consistency indicates that the model has learned effective representations that group same-class samples together while being robust to noisy labels. We use manifold neighborhood consistency as the evaluation metric, calculated by $\left(\frac{1}{n}\sum_{i=1}^{n}\frac{c_i}{k}\right)\times 100\%$, where n is the total number of training samples, k is the number of neighbors in the local manifold, and c_i represents the number of neighbors whose labels match the central sample's label. By tracking this metric across varying noise levels (0.1–0.4), we assess the model's ability to learn discriminative representations that respect the underlying data distribution despite noisy labels. Fig. 11 shows the evolution of manifold neighborhood consistency (with $k = 10$) over 200 training epochs. As noise increases, accuracy initially decreases but recovers over time, eventually exceeding 93% on the KSC, KC, and UP datasets. This sustained improvement demonstrates the effectiveness of our NTCRL CL strategy, which enhances category consistency within the local manifold and reduces the impact of noisy labels by guiding the model to focus on more reliable semantic relationships.

K. Effectiveness of NTCRL

We analyze the contribution of the CL module in the MACCL framework. Fig. 12 presents the experimental results on the KSC and UP datasets with noise rates of 0.2 and 0.4, respectively. We compare our CL method, NTCRL, with existing CRL techniques, including SimCLR [54] and SupCon [26]. For the CL method in (9), we represent it as CACRL. SimCLR is a self-supervised CL method, while SupCon is a supervised CL method. It is noteworthy that SimCLR performs the worst due to the conflict between the learning objectives and the supervision loss. Although SupCon performs better, it still struggles with noisy labels, as the sample pair selection based on noisy labels interferes with the entire representation learning process, making it difficult to learn effective spectral representations. In contrast, CACRL selects relatively clean pairs based on confidence, but as the noise rate increases, noisy pairs dominate the representation learning, making it difficult to obtain discriminative spectral representations. Finally, NTCRL effectively guides noisy samples in the correct semantic direction, avoids the gradient influence of incorrect pairs, and prevents overfitting to clean data, thus producing more discriminative representations.

L. Complexity Analysis

We analyzed the complexity of the proposed MACCL model in terms of model size, training time, and inference time under a 0.4 noise rate in the KSC dataset. First, Table IX shows the model size of MACCL and compares it with deep

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TABLE IX
MODEL SIZE, TRAINING TIME, AND INFERENCE TIME OF EACH ALGORITHM ON THE KSC DATASET

Methods	Model Size (MB)	Training Time (s)	Inference Time (s)
DCRN	2.71	**27.66**	1.59
TCRL	**0.60**	181.12	3.71
MASS	1.21	103.61	1.11
SPTNet-NTLA	1.48	289.63	1.08
GLHLE	0.92	256.06	2.21
MACCL	0.68	229.74	**0.86**

learning-based models. The results indicate that the MACCL model size is only 0.68 MB, making it one of the lightest models, thanks to the use of the lightweight TCRL architecture, which significantly reduces the number of parameters. Second, regarding the training time analysis, SPTNet-NTLA is the most time-consuming model due to its use of a co-training framework, while DCRN is the fastest method in terms of training time. Although MACCL's training time is slightly longer than most models, mainly because its core manifold-aligned consistency learning module requires additional manifold neighborhood prediction information. Finally, it is worth noting that after training with the noise-robust learning algorithm, MACCL achieves the shortest inference time. These results suggest that MACCL is not only efficient in the noisy HSI classification task but also maintains relatively low complexity.

V. CONCLUSION

In this study, we introduce the MACCL framework for mitigating the adverse effects of noisy labels in HIC. The framework features a manifold-aligned consistency learning module that reduces confirmation bias by aligning predictions with the intrinsic data manifold, ensuring decision boundaries that faithfully reflect the original spectral–spatial distribution. In addition, the noise-tolerant CL module refines the spectral–spatial representation space by focusing on high-confidence sample pairs and enforcing a regularization that amplifies gradients from these reliable pairs over those from erroneous ones. Extensive experiments on several benchmark datasets show that MACCL improves classification accuracy and remains robust under varying levels of label noise.

Pros and Cons: Pros integrates manifold learning with contrastive representation learning to combat noisy labels, delivering consistently strong results, and produces smoother, more robust decision boundaries than competing methods, remaining reliable even under heavy label noise. Cons requires additional manifold neighborhood prediction information, which introduces a modest increase in training time compared with methods that forgo such information.

Limitation and Future Work: Despite these advantages, some limitations remain. For instance, the proposed method does not account for the correlation between data noise and noisy labels, as real-world noisy labels may arise from inherent noise in the hyperspectral data itself, which can lead to errors in both automated and manual labeling. Future research will focus on addressing the challenges of noisy labels caused by sample degradation, aiming to develop associative learning models that can dynamically correct mislabels due to quality variations in HSIs. This will enhance the practical applicability of HIC in real-world scenarios.

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